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Piezoelectricity of strongly polarized ferroelectrics in prototropic organic crystals†

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The replacement of lead-containing piezoceramics is increasingly in demand for environmentally benign applications in various electromechanical devices. However, small organic molecules have attracted little interest in relation to their piezoelectric properties. We examined the direct and the converse piezoelectric effects at room temperature of several prototropic organic ferroelectrics having large spontaneous polarizations. The magnitude of the piezoelectric coefficient d_{33} showed a positive correlation with the spontaneous polarization. Whereas the maximum d_{33} ($\sim$15 pC N^{-1} for croconic acid) is comparable to the highest level ever observed among nonferroelectric small molecules, it is about half that of ferroelectric polymers and one or two orders of magnitude smaller than those of the commercially available piezoceramics. The modest piezoelectricity of prototropic ferroelectrics, despite their strong polarization, is opposite in behavior to that of ferroelectric Rochelle salt or triglycine sulfate. Plausibly, its origin can be ascribed to thermally robust ferroelectricity and a microscopic mechanism of ferroelectric switching that minimizes molecular deformations and reorientations.

Introduction

Piezoelectricity is the generation of electricity in a substance in response to an applied mechanical stress (the direct effect); it also produces mechanical strain in response to the presence of an external electric field (the converse effect). Such mechanical–electrical interconversion has long been utilized in diverse practical and potential applications, including sensors, energy harvesters, ultrasonic imaging, sonar, micro/macro actuators, micro-positioners, inkjet printers, and loudspeakers.[1–4]

Piezoelectricity occurs in a limited range of classes of crystals; only the 20 noncentrosymmetric point groups of the 32 classes of crystal are piezoelectric. Among the piezoelectric crystals, especially strong electromechanical effects appear in a group of polar substances known as ferroelectric substances, in which an external electric field can reversibly switch the direction of spontaneous polarization.[1,5,6] Unfortunately, many commercial piezoelectric devices still include ferroelectric ceramics that contain highly toxic lead, for example, zirconate titanate (PZT) and its modified derivatives, which have high piezoelectric coefficients. Consequently, many efforts have been dedicated to developing lead-free alternatives such as organic ferroelectrics.[4,7]

In recent decades, there have been many intriguing discoveries of strong polarization, low-field switching, and above-room-temperature ferroelectricity in small-molecule organic compounds.[8–12] The observed coercive field is low, typically ranging from 0.5 to 50 kV cm^{-1}, which is promising for low-voltage operation, unlike ferroelectric polymers,[4,13] for which the corresponding values exceed several hundreds of kV cm^{-1}. The ferroelectric phase is often sufficiently thermally robust to permit practical use at room temperature, because the Curie point is high and can even exceed the limits of thermal stability of the substance itself (its fusion, sublimation, or decomposition temperature). Importantly, the magnitude of the spontaneous polarization is comparable to or exceeds those of ferroelectric polymers ($\sim$8 μC cm^{-2}). The spontaneous polarization of $\sim$30 μC cm^{-2} at room temperature in croconic acid (**CRCA**; 4,5-dihydroxycyclopent-4-ene-1,2,3-trione) is the highest known value for any organic system, and it is comparable to typical values for lead-containing ferroelectrics.[11] Nevertheless, the piezoelectricity of small molecules remains relatively unexplored.

Here, we report the piezoelectric coefficients of five strongly polarized organic ferroelectrics: **CRCA**,[8,11] phenylmalonaldehyde (**PhMDA**),[9] 3-hydroxy-1*H*-phenalen-1-one (**HPLN**),[9] 2-methyl-1*H*-benzimidazole (**MBI**),[10] and 5,6-dichloro-2-methyl-1*H*-benzimidazole (**DC-MBI**).[10] The ferroelectricity of these compounds originates

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from a cooperative prototropy (also called 'proton tautomerism'), wherein a hydrogen atom is relocated and, simultaneously, the location of a single bond and that of an adjacent double bond are interchanged. Recently, extension of the field amplitude with the aid of an insulating oil has successfully permitted us to optimize the switchable polarizations up to the theoretical level by sweeping out pinned ferroelectric domains.[11] With the fully polarized crystals in hand, we can evaluate their intrinsic piezoelectric properties. For **CRCA** crystals, we compared the response with values of the field-induced change in lattice parameters, as recently determined by X-ray diffraction.[14]

Experimental

Preparation

All chemicals were purchased from commercial sources. Purification and crystal growth of orange rectangular plates of **CRCA** were performed by repeated slow evaporation of a solution in 1 M aqueous HCl. Colorless rod crystals of purified **PhMDA** and colorless plates of **MBI** and **DC-MBI** were grown by sublimation under reduced pressure. Orange plates of **HPLN** (α-form) were grown from a solution in ethanol. Triglycine sulfate (TGS) was prepared from the stoichiometric mixture of glycine and sulfuric acid and recrystallized from aqueous solution.

Electric poling and polarization measurements

All the electric and electromechanical measurements were performed on single crystals with painted silver electrodes. The two ends of the rod were cut along the (001) plane to accommodate the electrodes on the **PhMDA** and **DC-MBI** crystals. The as-grown crystals were used for the other compounds. The electric polarization–electric field (P–E) hysteresis curves were recorded on a ferroelectrics evaluation system (FCE-1; Toyo Corp., Japan) that consisted of a current/charge–voltage converter (model 6252), an arbitrary waveform generator (Biomation 2414B), an analogue-to-digital converter (WaveBook 516), and a voltage amplifier (HVA4321; NF Corp., Yokohama, Japan). The measurements at room temperature were performed with a high-voltage triangular wave field at various alternating frequencies. The crystals were immersed in silicone oil to prevent atmospheric discharge under the high electric field (>30 kV cm^{-1}) during the P–E hysteresis measurements, including the poling procedures. The achievement of an optimum piezoelectric response demands electric poling into a single-domain state. The P–E hysteresis loops were optimized with electric and/or thermal pretreatments, according to our recent report,[11] until the remanent polarization P_r reached an optimum level that almost coincided with the theoretical spontaneous polarization. The **CRCA** crystals were repeatedly exposed to a bipolar high-voltage rectangular wave field until the switchable polarization $2P_r$ reached the theoretical value[11] ($\sim$60 μC cm^{-2}). The α-form **HPLN** crystal was optimized by repetitive switching after heating at $\sim$420 K, whereas the hard ferroelectric **DC-MBI** crystal was fully poled by reducing the switching field, with the aid of heating to $\sim$360 K. The **PhMDA** and **MBI** crystals exhibited maximal values of $2P_r$ in their pristine states because they were grown at sufficiently high temperatures.

Electromechanical measurements

The piezoelectric coefficient d_{33} of the direct effect was measured by the Berlincourt method[15] using a piezometer system (PM300, PiezoTest Ltd, UK), which was calibrated with a PZT disk of known d_{33}. The poled specimen was clamped at its two electrode-painted crystal surfaces and a constant force of 0.3–1.0 N was applied. The piezoelectric charge resulting from the action of an additional dynamic force of 0.05–0.10 N (rms) oscillating at a frequency of 110 Hz was then measured. In comparison with the standard test conditions for piezoceramics, these static and dynamic forces were kept at modest levels to prevent mechanical damage to the samples under testing. The observed piezoelectric coefficients were validated by their independence of various static forces. The intrinsic piezoelectricity was also confirmed by the reversed polarity of the piezoelectric-charge response of the poled specimen mounted upside down. The magnitude of d_{33} was averaged over the measurements with the regular and reversed configurations.

The piezoelectric coefficient d_{33} of the converse effect was determined by measuring the longitudinal strain, which was synchronized with the P–E hysteresis experiments by additionally equipping the FCE-1 ferroelectrics evaluation system described above with a heterodyne interferometer (LV-2100; Ono Sokki, Yokohama). The fully poled single crystal was completely immersed in insulating silicone oil and sandwiched between the bottom electrode and a thin aluminum-rod electrode (3 mm in diameter, 4 cm in length, weighing 1.29 g, with a 0.5 mm-diameter tip) that was placed on it (Fig. 1a). This experiment required crystals of a sufficient size, especially in terms of the areas of the flat surfaces for the two electrodes. The dimensions of the **CRCA** crystal were $2.35 \times 1.52 \times 0.93$ mm^3 in the crystallographic b, a, and c directions, respectively, whereas the **PhMDA** crystal was 1.20 mm in length in the crystallographic c direction, with a section area of 1.40×0.70 mm^2. The **HPLN** crystal used had a well-developed surface (1.25×1.10 mm^2) parallel to the (10−1) plane, and a thickness of 0.60 mm. The vertical displacement Δl of the top electrode surface on the crystal was monitored by reflecting an incident He–Ne laser beam from the top mirror of the rod electrode. The strain was obtained as $x_3 = \Delta l/l$, where l is the crystal length (interval between the two electrodes). The interferometer and the sample stage were located on an active vibration-isolation table. To minimize noise levels in the strain, the data were continuously collected at a frequency of 50–100 Hz and averaged over 100 cycles.

The piezoelectric resonance measurements were performed by measuring the impedance $|Z|$ and phase angle as functions of the frequency on a Solartron impedance gain-phase analyzer 1260 and a dielectric interface 1296. This experiment demanded crystals of sufficiently anisotropic dimensions to permit activation of the specific vibrational mode. A single-crystal rectangular plate of **CRCA** was poled along its thickness direction and employed as a resonator with a length-extensional (LE) mode. The dimensions of the crystal were l = 4.1 mm, w = 1.4 mm, and t = 0.70 mm in the crystallographic b, a, and c directions, respectively.

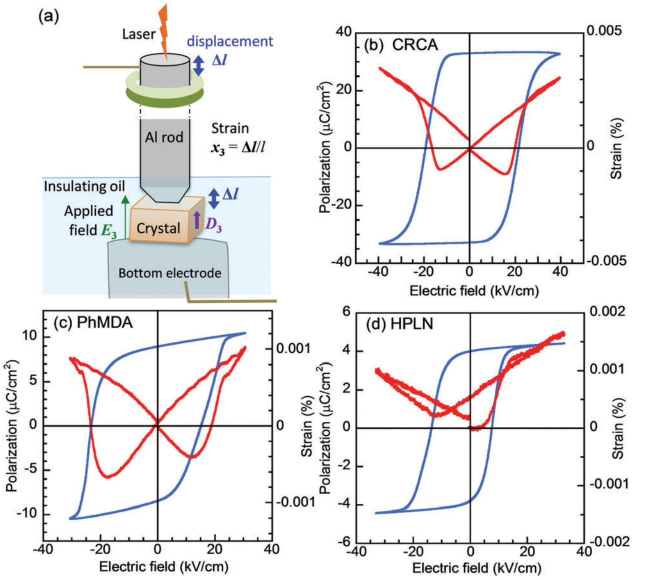

Fig. 1 Converse piezoelectricity of several prototropic ferroelectric crystals. (a) Experimental setup. (b–d) Electric polarization (P) *versus* electric field (E) hysteresis loops and simultaneously E-dependent strains measured with E applied (b) along the c-direction of the **CRCA** crystal at f = 50 Hz, (c) along the c-direction of the **PhMDA** crystal at 50 Hz, and (d) normal to the (10−1) plane of the **HPLN** crystal at 100 Hz.

Results and discussion

Piezoelectric assessment

The direct and converse effects in piezoelectric crystals are described by the coupling between elastic variables (stress X and strain x) and electric variables (electric field E and dielectric displacement D or polarization P) through the piezoelectric coefficients. In this study, we examined the piezoelectric d coefficients of prototropic ferroelectrics. The corresponding basic equations of state are

$$D = dX + \varepsilon^X E \quad (1)$$

$$x = s^E X + {}^t dE \quad (2)$$

where ε is the dielectric permittivity, s is the material compliance, and superscripts E and X indicate the variables held constant.[1–3] For the actual piezoelectrics, E, D, and x are directional quantities, and the parameters d, ε, and s are in a tensor form and superscript t represents the transpose. The subscripts of the matrix elements designate the experimental conditions. For instance, the coefficient d_{33} (or d_{31}) in the direct effect is related to the polarization generated along direction 3 in response to a stress applied in the longitudinal direction 3 (or the vertical direction 1) through the following relationship:

$$D_3 = d_{33}X_3 \text{ (or } D_3 = d_{31}X_1) \quad (3)$$

The converse effect is described by the same coefficient d_{33} (or d_{31}) through the similarly simple form

$$x_3 = d_{33}E_3 \text{ (or } x_1 = d_{31}E_3) \quad (4)$$

The ferroelectric **CRCA** and **DC-MBI** crystals adopt the orthorhombic space group $Pca2_1$, whereas **PhMDA** adopts $Pna2_1$; all three crystals belong to the same point group, $mm2$.[8–10] The nonzero matrix elements of the piezoelectric coefficient d are d_{31}, d_{32}, d_{33}, d_{15}, and d_{24}, where the Cartesian axes, 1, 2, and 3 are parallel to the crystallographic a, b, and c axes, respectively. Note that the sign of d was defined with a positive polarization in the +z direction. Here, we consider the 'normal' stress (compression or tension) and the strain perpendicular to the surface, but not the 'shear' components of stress and strain related to subscripts 4, 5, and 6, defined according to the so-called contracted Voigt subscript notation. d_{33} represents the piezoelectricity of the longitudinal mode, whereas d_{31} and d_{32} represent that of the transverse mode. **HPLN** and **MBI** exhibit a lower crystal symmetry as they belong to space group (point group) Pn (m).[9,10] For convenience, the +z direction in the Cartesian coordinate system was similarly set in the direction of the electric poling (and the applied stress), which deviated from all the crystallographic a, b, and c directions. Thereby, their piezoelectric coefficients can also be designated as d_{33}. See Fig. S1 (ESI†) for the crystallographic directions on the single crystals.

Direct effect

Before the piezoelectric measurements were performed, all the crystals were electrically poled with an external field (in the +z direction). Because the evaluation of intrinsic piezoelectric properties demands a fully polarized (single-domain) state of the crystal, the magnitude of the switchable polarizations in the P–E curves were improved up to the optimum level (Fig. S2, ESI†) by repeated switching with a high ac electric field. The piezoelectric coefficient d_{33} of the direct effect, described by the following equation,

$$d_{33} = \left(\frac{\partial D_3}{\partial X_3}\right)_E \quad (5)$$

was measured by using a piezometer under short-circuit conditions (constant E = 0). The piezoelectricity of all the compounds was evidenced by observations of modest but nonzero d_{33} values, as summarized in Table 1.

Converse effect

The simultaneous measurements of polarization and strain revealed the converse piezoelectric effects associated with ferroelectricity (Fig. 1). For all the prototropic ferroelectrics examined, the longitudinal strain of the negatively poled crystal decreased linearly with increasing field from zero bias. Beyond the positive coercive field, the crystal was switched to a positively polarized state and expanded rapidly from the minimum to maximum strain. The crystal exhibited a linear contraction again on decreasing the field until it switched back to the negatively polarized state. The strain–electric field curve therefore revealed butterfly loops around the pair of coercive fields, as is typical for the piezoelectricity of ferroelectric materials. By applying eqn (4), the piezoelectric coefficient d_{33} could be determined from the slope of the linear strain effect at about zero field. For each crystal, the d_{33} values for the converse effect were comparable to

Table 1 Piezoelectric strain d coefficients of organic crystals at room temperature. Top (1–5): this work on the prototropic ferroelectric crystals. Bottom: reported data on single-component small-molecule piezoelectric molecules and some selected room-temperature ferroelectric salts containing small molecules

Compound[a]	T_c or T_d[b] (K)	Longitudinal mode[c,h] Direct (pC N^{-1})	Longitudinal mode[c,h] Converse (pm V^{-1})	Notation	Definition of axes (1, 2, 3)[g]	Transverse mode[c,h] (pm V^{-1})		Point group	Ref.
1. **CRCA**	>420	+15.1(10)	+7.6(5)[e]	d_{33}	(a, b, c)	d_{32} = 8.0[d]		$mm2$	
2. **PhMDA**	363	+10.4(10)	+4.3(2)[e]	d_{33}	(a, b, c)			$mm2$	
3. **DC-MBI**	399	+12.2(3)		d_{33}	(a, b, c)			$mm2$	
4. **MBI**	>430	+7.5(4)		d_{33}	axis3 ∥ [101]			m	
5. **HPLN**	>420	+7.3(8)	+3.1(8)[e]	d_{33}	axis3 ⊥ (10−1) plane			m	
Benzil	—	6		d_{33}	(a, b*, c)			32	17
FMA	—		8.2(4)[e]	d_{33}	(a, b, c)	d_{31} = 2.2(2)[e]		4mm	18
Resorcinol	—	+5.6		d_{33}	(a, b, c)	d_{31} = −4.1	d_{32} = −4.3	$mm2$	19
mNA	—	6.8		d_{33}	(a, b, c)	d_{31} = 30.8	d_{32} = 2.6	$mm2$	20
MNA	—		13.8[f]	d_{11}	(a, b, c*)	d_{12} = 2.5[f]	d_{13} = 1.4[f]	m	21
Anthracene	—	+0.15		d_{22}	(a, b, c*)	d_{21} = −0.07	d_{23} = −0.08	2	22
MBANP	—	14.9		d_{22}	(a, b, c*)	d_{21} = 16.1	d_{23} = 0.4	2	23
Rochelle salt	297	−33	−50	d_{22}	(a, b, c*)	d_{21} = 11	d_{23} = 21	2	24 and 25
TGS	323	18.9, +29.1(7)[i]	20[e]	d_{22}	(a, b, c*)	d_{23} = 15.3		2	26 and 27
		7.9[d], +32.7[d]		d_{22}	(a*, b, c)	d_{21} = 23.6[d], +20.8[d]	d_{23} = 25.3[d], −61.2[d]	2	28 and 29

[a] FMA = 2-furylmethacrylic anhydride; resorcinol = 1,2-$C_6H_4(OH)_2$; mNA = m-nitroaniline; MNA = 2-methyl-4-nitroaniline; MBANP = (−)-2-(α-methylbenzyl)amino-5-nitropyridine; TGS = triglycine sulfate. [b] Curie temperature T_c or depolarization temperature T_d was determined in the previous work on the thermal and dielectric properties (ref. 8–11). [c] Data derived from the direct stress–charge method unless noted. [d] The resonance–antiresonance method. [e] Macroscopic strain measurements. [f] Microscopic strain measurements from X-ray diffraction. [g] The coordinate axis 3 refers to the poling direction for the definition of d_{33} of the prototropic ferroelectrics. [h] Only absolute values were available for the d coefficients without signs. [i] The largest d_{22} was found in this work using the specimen with fully optimized P_r (2.8 μC cm^{-2}), which is much higher than P_r (1.19 μC cm^{-2}) of ref. 26 but close to the spontaneous polarization[16] (3.0 μC cm^{-2}).

those for the direct effect (Table 1; note the relationship between the corresponding units, pm V^{-1} = pC N^{-1}).

Piezoelectric resonance

The observation of a modest d_{33}, as described above, despite the strongest polarization of the **CRCA** crystal, qualitatively agrees with the recent results of X-ray diffraction studies conducted by Kobayashi *et al.*[14] Moreover, we also examined the piezoelectric resonance–antiresonance appearing in the impedance spectra. For the as-grown **CRCA** crystals, one pristine crystal had dimensions that permitted it to act as a resonator with a length-extensional (LE) mode. When the ac electric field was applied along axis 3, the vibration direction was parallel to the 2 axis (Fig. 2a, inset), permitting its resonance frequencies to be related to the piezoelectric constant d_{32}. Fig. 2 shows the variations in the impedance $|Z|$ and the phase angle as functions of the frequency. A sharp dip and peak in $|Z|$ at a frequency of about 4 MHz were accompanied by a peak in the phase angle and they are characteristic of piezoelectric resonance and antiresonance. The relative permittivity at a constant stress $\varepsilon_{33}^X/\varepsilon_0$ was calculated to be 13.2 from the capacitance C_f = 0.96 pF at ∼10 kHz and the permittivity of a vacuum of ε_0 = 8.8542 × 10^{-12} F m^{-1}, together with the sample dimensions l, w, and t noted above, was calculated by using the relationship below:

$$\frac{\varepsilon_{33}^X}{\varepsilon_0} = \frac{C_f t}{\varepsilon_0 w l} \tag{6}$$

The present data could not be satisfactorily analyzed with the equivalent-circuit model, because the resonance was not strong enough to switch the phase angle beyond 0°. Instead, the dip and

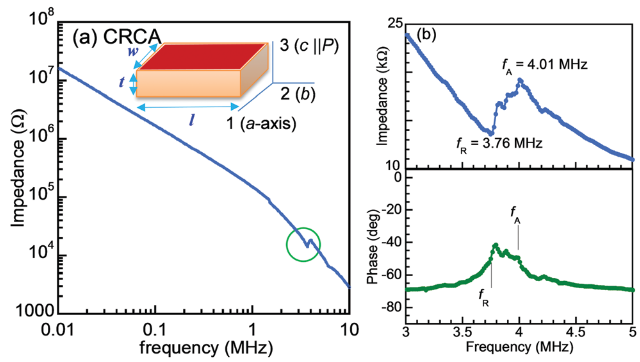

Fig. 2 Impedance *vs.* frequency curve of croconic acid (**CRCA**) at room temperature. The arrows indicate the resonance and antiresonance frequencies. The inset to panel (a) shows the configuration for the measurements.

peak frequencies, which were well defined and separated from each other, were assumed to approximate the frequencies of resonance and antiresonance, respectively, to permit an estimation of the piezoelectric coefficients. The elastic compliance at a constant electric field s_{22}^E, as well as the velocity of sound v, can be readily calculated from the resonance frequency f_R by the following relationships:

$$f_R = \frac{v}{2l} \tag{7}$$

and

$$v = \frac{1}{\sqrt{\rho s_{22}^E}}, \tag{8}$$

where ρ is the density of the crystal. The observed resonance frequency f_R = 3.76 MHz at the dip and ρ = 1912 kg m^{-3} (available from the X-ray diffraction data) gave $\nu = 1.196 \times 10^4$ m s^{-1} and $s^E_{22} = 3.656 \times 10^{-12}$ m^2 N^{-1}. The observed antiresonance frequency f_A = 4.01 MHz at the peak (f_A/f_R = 1.0665) gives the following relationship:

$$\frac{k_{32}{}^2}{1-k_{32}{}^2} = -\frac{\pi f_A}{2 f_R}\cot\frac{\pi f_A}{2 f_R} = 0.1756 \tag{9}$$

from which we can determine the electromechanical coupling factor

$$k_{32} = \frac{d_{32}}{\sqrt{s^E_{22}\varepsilon^X_{33}}} = (\pm)0.3865, \tag{10}$$

as well as the piezoelectric strain coefficient d_{32} = (±)7.99 pm V^{-1}.

Note that the magnitudes of the experimentally determined d_{32} and d_{33} are comparable to one another, as is often the case for ferroelectrics.

Comparison of piezoelectric and ferroelectric properties

Among the three methods that we employed, only measurements of the direct piezoelectric effect permitted the determination of d_{33} for all five compounds. On the other hand, the converse effect was successfully examined for only three compounds. The slightly smaller values of d_{33} might be due to the effect of domain-wall motions. In Fig. 3, it is possible to perceive a positive correlation between the spontaneous polarization P_s and the values of d_{33} for the direct effect. Table 1 lists the d coefficients reported for single-component small-molecule piezoelectrics[16–23] and for several selected room-temperature ferroelectric salts [Rochelle salt[24,25] and TGS[26–30]] that consist of small molecules (Chart 1). In contrast to the ferroelectric and piezoelectric polymers,[4,13,31–34] few small molecules have had their piezoelectric properties examined. Because piezoelectricity is accompanied by a second-harmonic generation activity, most of the reported studies involved nonferroelectric organic compounds that exhibited strong optical nonlinearity.[35] The largest coefficient that we observed ($d_{33} \approx$ 15 pC N^{-1} for **CRCA**) is comparable to those of 2-methyl-4-nitroaniline (MNA)[21] and of (−)-2-α-methylbenzylamino-5-nitropyridine [MBANP; 5-nitro-2-(1-phenylethyl)pyridine],[23] which were each claimed to be the highest ever observed among small-molecule crystals. Likewise, the **CRCA** crystal exhibited a strong second-harmonic-generation activity.[36] On the other hand, Fig. 3 reveals that even the d_{33} of **CRCA** is about half those of poly(vinylidene fluoride) (PVDF) (12–30 pC N^{-1})[32] or poly[(vinylidene fluoride)-*co*-trifluoroethylene] (P(VDF-TrFE)) (30–42 pC N^{-1}),[33,34] and one or two orders of magnitude smaller than those of commercially available PZT-based piezoceramics (100–800 pC N^{-1}). The modest piezoelectricity of the prototropic ferroelectrics, despite their strong polarization, is opposite in behavior to that of ferroelectric Rochelle salt or TGS. One reason for this dissimilarity might be the distinct thermal stability of the ferroelectric state. The electromechanical properties can be enhanced by inducing structural instabilities, such as morphotropic phase boundaries or thermally induced phase transitions.[7] An approach toward the Curie temperature (T_c) from below generally reduces polarization but enhances the piezoelectric coefficients for ferroelectrics.[30] Both the ferroelectric salts have values of T_c immediately above room temperature, in sharp contrast to the thermally robust ferroelectricity of the prototropic compounds (Table 1). Another plausible reason involves the mechanism of ferroelectric switching. The prototropy might reverse the polarization without a significant change in the molecular shape and orientation, thereby minimizing the resultant strains and d_{33}.

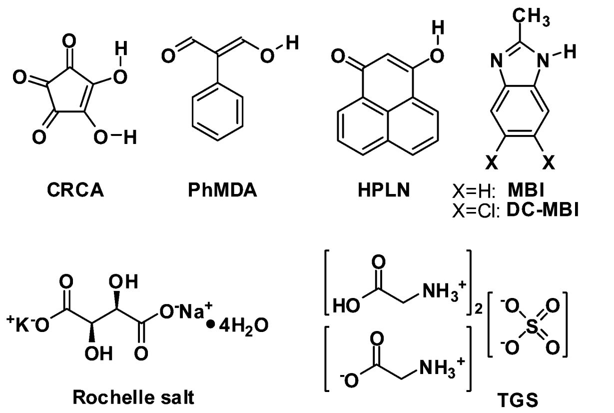

Chart 1

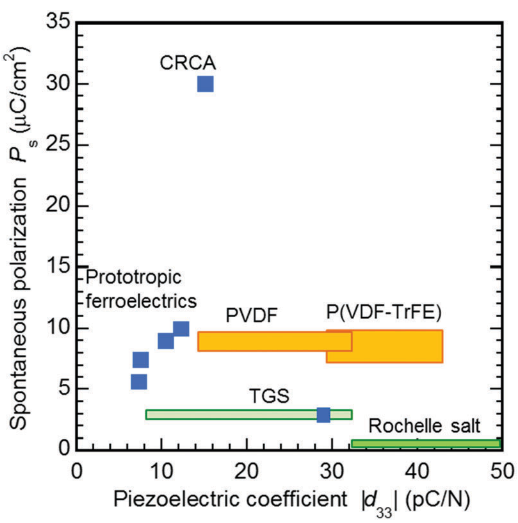

Fig. 3 Relationship between the spontaneous polarizations and the longitudinal piezoelectric coefficients d_{ii} (i = 1, 2, or 3) at room temperature for the ferroelectric prototropic crystals in comparison with the reported properties of TGS, Rochelle salt, and ferroelectric polymers. Blue squares on prototropic and TGS crystals represent the data derived from the direct stress–charge method in this work.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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